

AIL + MoIE

Recursive Research Re-Entry Protocol

A production-ready specification for evidence-governed, AI-assisted research and engineering

Core principle: Experiments do not merely generate results. They generate structured evidence that re-enters the reasoning system, updates the active theory under uncertainty, and selects the next discriminating test.

Governing rule: *The experiment decides. The evidence updates the system. The updated system asks the next question.*

1. Purpose and scope

This protocol governs an iterative research loop for scientific, technical, product, operational, and AI-system investigations. It is designed to prevent premature conclusions, preserve evidence lineage, compare competing mechanisms, and turn each completed experiment into a justified next action.

It does not guarantee truth or eliminate expert judgment. It creates a disciplined process for revising claims in proportion to traceable evidence.

2. Complete research loop

Problem → Axioms and assumptions → Inversion → Competing mechanisms → Hypothesis → Prediction → Experiment design → Execution → Results → Evidence synthesis → JEV analysis → Theory update → Next hypothesis / test

A research cycle is complete only when the evidence package has been preserved, adjudicated against the active model, and used to produce either a new experiment, a decision, a stop condition, or an escalation to human review.

3. Non-negotiable rules

- Observation, derived result, interpretation, and causal conclusion must be recorded as separate statements.
- No causal claim may be promoted solely because an observation is consistent with it. Competing mechanisms and confounds must be assessed.
- Every major claim must point to a source dataset, experiment identifier, analysis method, and execution environment.
- A failed hypothesis is retained as evidence; a supported hypothesis remains provisional and scoped to tested conditions.
- No result is called robust until its reproducibility status is explicitly recorded.
- The system must report uncertainty, missing data, and residual risk rather than silently filling gaps with narrative.
- High-stakes decisions require a named human approver and an auditable decision record.

4. Research state model

Object	Required contents
Problem	Decision context, stakeholder, target outcome, constraints, harm boundaries, and success criterion.
Assumption	Statement, rationale, confidence, impact if wrong, testability, owner, and status.
Mechanism	Candidate causal or operational explanation, prerequisites, predicted signatures, and alternatives.
Hypothesis	Falsifiable claim with scope, variables, direction, measurable outcome, and boundary conditions.
Prediction	Pre-registered expected result, primary metric, threshold, analysis plan, and stopping rule.
Experiment	Design, controls, sample or coverage plan, instrumentation, risks, environment, and reproducibility package.
Evidence	Raw artifacts, validated measurements, derived effects, uncertainty, provenance, and quality assessment.
Theory	Current best explanation, supporting and contradictory evidence, limitations, confidence, and unresolved questions.

5. Evidence package

Every completed experiment produces an evidence package before interpretation is allowed. The package has four layers.

Evidence summary

One concise paragraph stating only the highest-value observed and derived findings. It contains no new theory, causal language, or unsupported explanation.

Research overview

A human-readable map of the problem, assumptions, inversion, mechanisms, hypothesis, prediction, design, major results, support or falsification status, and limitations.

Detailed evidence breakdown

- Finding
- Measurement and unit
- Comparison or baseline
- Effect size
- Uncertainty and confidence interval where applicable
- Supporting evidence
- Contradicting evidence
- Potential confounds
- Data-quality issues
- What remains unknown

Evidence provenance

Raw data references, code version or commit, dependency versions, container image digest, configuration fingerprint, operator identity, execution time, hardware or runtime context, and reproducibility instructions.

6. Typed evidence record

Use a structured record alongside prose so results can be reviewed, compared, and re-entered programmatically.

finding_id: F-001
hypothesis_id: H-001
experiment_id: E-001
claim_tested: "A increases B under condition C"
primary_metric: "B"
measurement_value:
measurement_unit:
baseline_value:
effect_size:
uncertainty:
sample_size_or_coverage:
analysis_method:
data_source:
data_quality_status: validated | incomplete | suspect
replication_status: unreplicated | internal | independent | robust
competing_mechanisms_considered: [M-001, M-002]
confounds_considered: [C-001, C-002]
jev_relation: EQUAL | NOT_EQUAL | PARTIAL | CONDITIONAL | CONFOUND | UNKNOWN | NOVEL
confidence: low | medium | high
limitations: []
provenance:

7. JEV evidence-relation analysis

JEV is the controlled routing layer between evidence and theory update. It classifies the relationship between observed evidence and the pre-specified prediction; it does not substitute for statistical analysis or expert judgment.

Relation	Meaning	Required next action
EQUAL	Observed evidence is consistent with the prediction within the tested scope.	Replicate and probe wider scope, sensitivity, and alternative mechanisms.
NOT EQUAL	Observed evidence contradicts the prediction or fails the pre-registered criterion.	Check design and measurement integrity; retain the failure; test alternatives.
PARTIAL	Only a subset of the predicted relationship appears.	Isolate the supported portion and design a test for the missing condition or mechanism.
CONDITIONAL	The relationship appears only under a defined boundary condition.	Test whether the condition is causal, necessary, sufficient, or merely correlated.
CONFOUND	An alternative factor could explain the observed result.	Add controls, randomization, stratification, instrumentation, or a negative control.
UNKNOWN	Evidence cannot distinguish among plausible explanations.	Improve measurement, coverage, power, or experiment discriminability.
NOVEL	The result reveals an unmodeled pattern or candidate mechanism.	Register the candidate; create a discriminating follow-up hypothesis.

8. Theory update protocol

A theory update may use only evidence that has passed provenance and quality review. The system must make the delta explicit rather than replacing the previous theory silently.

- Original theory: the pre-experiment explanation and its scope
- Observed evidence: findings with identifiers and quality status
- What changed: exact proposition(s) strengthened, weakened, removed, or added
- Competing mechanisms: which alternatives survived, weakened, or emerged
- New mechanism or boundary condition: stated as provisional
- Remaining uncertainty: unresolved variables and confounds
- New hypothesis: a falsifiable claim generated from the update
- Next test: the experiment most likely to discriminate among remaining explanations

9. Experiment selection

Choose the next experiment for information value—not merely convenience. Prefer the feasible experiment whose outcomes would most change the decision, reduce uncertainty between competing mechanisms, or expose a high-severity risk.

Selection score

Score candidate experiments on expected discriminating value, decision impact, feasibility, cost, time, safety risk, reversibility, data quality, and operational burden. Escalate to human review when potential harm or irreversibility exceeds the authorized threshold.

10. Quality gates

- Prediction locked before execution, including primary metric, threshold, and stopping rule
- Instrumentation validated before data collection
- Raw evidence preserved with immutable references
- Analysis reproducible from raw inputs
- At least one plausible alternative mechanism explicitly evaluated
- Confounds and negative results recorded
- Uncertainty quantified or qualitatively rated with rationale
- Results separated from interpretation
- Replication status recorded
- Human review completed for high-impact findings or decisions

11. Stop, escalate, or continue

Condition	Action
Decision convergence: further testing is unlikely to change the decision materially.	Document decision, residual uncertainty, scope, and monitoring plan.
Budget, time, sample, compute, or operational limit reached.	Stop; record evidence limitation and propose the highest-value future test.
Safety, ethical, legal, or security risk exceeds authority.	Stop execution and escalate to named human owner.

Condition	Action
Evidence remains non-discriminating after planned tests.	Declare UNKNOWN; redesign measurement or revise mechanism space.
New evidence invalidates a critical premise.	Re-enter at assumptions/inversion; do not continue as if the premise still holds.

12. Research memory and audit trail

Maintain a longitudinal, append-only record. Never overwrite an earlier theory, failed test, or discarded mechanism; supersede it with a linked update.

- Problem and decision context
- Assumptions and inversion records
- Mechanism graph and competing explanations
- Hypotheses and pre-registered predictions
- Experiment protocols and execution logs
- Raw evidence and processed evidence
- Evidence summaries and JEV classifications
- Theory versions and rationale for each delta
- Replication history and robustness status
- Risk register, approvals, escalations, and final decisions

13. Minimum artifact set

- Research brief: problem, scope, stakeholder, decision, and constraints
- Assumption and inversion register
- Mechanism map
- Hypothesis and prediction ledger
- Experiment protocol and pre-execution checklist
- Evidence package and provenance manifest
- JEV adjudication record
- Theory-update record
- Next-test decision memo
- Final white paper: evidence summary, research overview, detailed breakdown, updated theory, and next test

14. Final operating statement

This protocol treats successful results as scoped candidates for refinement and failed hypotheses as durable evidence. Its objective is not to defend an original idea. Its objective is to improve the explanatory and decision model after every completed test.

Evidence → theory update → next hypothesis → next discriminating experiment.